

# PAVANI RONANKI

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## SUMMARY

Software engineer with 3+ years of experience delivering scalable solutions using **JavaScript, React.js, Node.js**, and cloud platforms such as **AWS and Azure**. Strengthened application security by implementing robust authentication and **access-control workflows**, reducing unauthorized access incidents by 95%. Integrated **OpenAI GPT LLMs** into automated feedback systems to generate personalized code review insights that improved grading efficiency and overall student engagement. AWS Certified Cloud Practitioner.

## EDUCATION

**Master of Science in Computer Science** GPA : 3.76/4.00

*Iowa State University*

Ames, USA

**Coursework:** Cloud Computing, Distributed Systems, Quantum Computing, Software Engineering

**Bachelor of Technology in Computer Science** GPA : 3.55/4.00

*Jawaharlal Nehru Technological University*

Kakinada, India

**Coursework:** Operating Systems, Database Management Systems, Computer Architecture, Computer Networks

## EXPERIENCE

**Software Engineer (Research Assistant)**

August 2024 - Present

*Iowa State University*

Ames, USA

- Architected and deployed automated Java grading pipeline using **TypeScript** and **Python scripting**, reducing manual grading time by 85% and supporting 500+ student submissions weekly
- Integrated **OpenAI GPT LLMs** into the feedback pipeline to generate personalized, context-aware code review comments, improving feedback quality and boosting student learning outcomes by 40%.
- Pioneered RESTful microservices for Pearson's (LMS) using **Node.js** and **Express**, implementing optimized caching, API gateway routing, and **SQL query tuning**.
- Structured **GraphQL schema** with field-level authorization and JWT, enabling secure access control for instructors, students, and proctors across Pearson's assessment platform
- Provisioned **Apache Kafka** message streaming architecture for asynchronous processing of code submissions, handling peak loads of 200+ concurrent evaluations without performance degradation
- Built event-driven backend workflows using **AWS Lambda** and **SQS**, automating long-running tasks and improving **fault tolerance** reducing system-level failures by 30% across production environments.
- Leveraged **Git** for collaborative development, managing feature branches, performing code reviews, and enforcing version control best practices to maintain clean, stable, and high-quality codebases.

**Software Engineer**

October 2022 - July 2023

*Infosys*

Hyderabad, India

- Delivered a supplier management and complaint-tracking platform using **Java Spring Boot** and **React.js**, supporting 2,000+ active users across dual authentication portals and enabling streamlined issue reporting and improved workflow visibility.
- Implemented **Spring Security** with **JWT** authentication, reducing unauthorized access incidents by 95% and ensuring secure data transmission for supplier workflows.
- Designed complex NoSQL database schemas (**Oracle, MongoDB, Redis**) with optimized indexing strategies, implementing **JDBC** and Spring Data JPA for efficient data persistence layers.
- Enhanced **Node.js** serverless functions on **AWS Lambda**, implementing efficient data processing pipelines that reduced daily ETL execution time by 50% while maintaining data integrity.
- Formulated **ETL workflows** on **GCP** using Cloud Dataflow and **BigQuery**, streamlining large-scale data ingestion and achieving sub-second analytical query performance across millions of records.
- Orchestrated **CI/CD pipelines** using **Jenkins** and **GitHub Actions** on **AWS**, reducing deployment time from 2 hours to 15 minutes and enabling daily production releases
- Refined **PostgreSQL** performance by refining complex **SQL queries** and configuring efficient connection pooling, improving application response times by 60% and supporting a 50% increase in concurrent user load during high-traffic periods.
- Mentored 3 junior engineers in **agile methodologies** and testing frameworks, fostering code quality while improving team sprint velocity by 25%

**Software Engineer**

Feb 2021 - October 2022

*Tata Consultancy Services*

Pune, India

- Developed comprehensive manufacturing management system (POTTS) with **React.js** frontend and **C#/.NET backend**, supporting 1,000+ daily active users across production tracking workflows.
- Built 10+ interactive **Power BI** dashboards to visualize customer business insights and track key performance metrics.
- Implemented workflows with Azure Data Storage Factory and **Azure Databricks** for large-scale data processing.

- Provisioned scalable cloud infrastructure on **Azure with Kubernetes** orchestration, delivering 99.5% system uptime and seamless handling of peak production workloads across distributed environments
- Strengthened resilient large-scale system components, improving **fault tolerance** by 25% and reducing operational costs through refined resource utilization across decentralized environments.
- Utilized **SSIS** (SQL Server Integration Services) and **PowerShell** scripts for efficient data migration and transformation.
- Customized Salesforce workflows, roles, and permissions for 50+ users, ensuring 100% **data security** compliance.
- Provided post-implementation support for 7+ applications, reducing downtime by 20% through intense **debugging**.
- Created unit and integration tests using **Jest and Selenium**, reducing UI defects and improving release stability by 30%
- Applied **SCRUM Agile methodology**, successfully delivering 6+ sprint cycles on time with 95% sprint completion rate.
- Created 12 detailed documentation and collaborated with teams to deliver business-driven data solutions.

## PROJECTS

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- **AI-Powered Data Analysis Platform:** Crafted an AI-powered analytics platform using C# .NET Core with GPT-based natural language querying, integrated Python data processing pipelines, optimized SQL Server schemas, and launched the system on Azure for scalable performance.
- **Distributed E-Commerce Platform:** Engineered microservices-based e-commerce system using Java Spring Boot with REST APIs, Provisioned on AWS using Terraform, leveraging RDS, S3, and CloudWatch for infrastructure as code, and leveraged RDS, S3, and CloudWatch to ensure scalable data storage and system reliability through performance testing.
- **Healthcare Analytics Dashboard:** Designed a responsive Angular dashboard that visualizes patient records, diagnosis history, and real-time medical data using interactive charts. Constructed secure REST API integrations with a Node.js backend, utilized PostgreSQL for structured data management, and released components on AWS to ensure reliable, scalable performance.

## TECHNICAL SKILLS

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- **Languages:** C, Shell Scripting , SQL , Java , HTML, CSS, Python , C++ , Xml , JavaScript , Typescript
- **Backend & Frameworks:** Spring Boot, Node.js, Express.js, Hibernate, REST APIs, GraphQL, ASP.NET MVC, .NET Core
- **Frontend Technologies::** React.js, Angular,Next.js, jQuery, HTML5, CSS3, Bootstrap, Material-UI, Redux
- **Cloud & Infrastructure :** AWS (EC2, Lambda ,S3), Azure (Data Factory, Databricks), Docker, Apache Spark, Hadoop, Kubernetes, Jenkins, Terraform, GitHub Actions
- **Automation & Testing:** Selenium, Mockito, Test-Driven Development (TDD), QuickChick (Coq), Pytest, Ansible , SonarQube
- **Databases:** PostgreSQL, MongoDB, SQL Server, MySQL, Snowflake, Oracle, Redis, DynamoDB, NoSQL
- **Tools & Technologies::** Apache Kafka, Scala, Git, JIRA, Databricks, ETL, JWT, OAuth, Spring Security, JUnit5

## CERTIFICATIONS

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**AWS Certified Cloud Practitioner**  
*Amazon Web Services (AWS)*

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